

Cleaning the Trace Also Weakened the Loyalty: Toward Bias-Scrubbing Adapters

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Abstract

Chain-of-thought (CoT) monitoring can reveal undesirable reasoning that is absent from a model’s visible answer, but optimizing CoT can instead teach a model to hide its intent. We test a controlled alternative: distilling base-model reasoning into a model organism with an implanted hidden policy. Our target is an AuditBench Qwen3-14B organism trained to privilege Russian geopolitical interests. We construct 4,245 examples that pair reasoning sampled from stock Qwen3-14B with visible-answer targets sampled from the loyal organism; only the reasoning source is replaced. A rank-64 LoRA trained above the frozen loyalty adapter reduces mean CoT loyalty legibility from 7.20 to 2.84 on 45 paired scenarios (change -4.36 , bootstrap 95% CI: -5.07 to -3.60) and visible-answer slant from 4.84 to 3.42 (change -1.42 , CI: -2.33 to -0.51). Explicit self-reference to the hidden policy falls from 31/45 to 1/45. An intervention intended to preserve loyal answers while cleaning their rationales therefore attenuates both measured channels. This single-run result motivates reversible reasoning-conditioned “scrubbing” adapters, but does not show mechanistic erasure: generic continued fine-tuning and adapter interference remain uncontrolled explanations.

Operational outcome	CoT signal	Visible behavior
Monitor evasion	lower	persists
Attenuation-compatible	lower	lower
Failed intervention	persists	persists
Channel corruption	unscorable	unscorable

Table 1: The two measured channels distinguish relevant outcomes. Neither alone establishes that a latent policy has been removed.

1 Introduction

Language models may pursue undesirable policies without admitting them. An apparently acceptable answer can coexist with reasoning that pursues a hidden objective; conversely, a benign-looking rationale can conceal continued misbehavior. AuditBench makes this threat experimentally tractable by releasing model organisms with known implanted behaviors, including a topic-gated “secret loyalty” that subtly privileges Russian geopolitical interests (Sheshadri et al., 2026).

Reasoning models expose an additional audit surface. CoT monitors can detect misbehavior that is difficult to infer from actions alone, but optimizing against those monitors can select for obfuscation (Baker et al., 2025; Haskins et al., 2026). Moreover, generated CoT need not faithfully report the computation producing an answer (Chen et al., 2025; Lanham et al., 2023). We therefore evaluate two channels independently: if suspicious CoT decreases while behavior persists, the intervention has concealed the signal; if both decrease, it is compatible with behavioral attenuation.

We originally tried to construct the first outcome. Each training target retained an answer from the loyal model while replacing only its rationale with reasoning from the stock base model. Contrary to our prior, visible answer bias weakened too. This is the central observation:

answer slant decreased even though every target answer came from the loyal organism.

Our contribution is not a claim to the first debiasing adapter. Instead, we provide:

- (1) a matched, dual-channel test that distinguishes concealment from attenuation;
- (2) a reversible counter-adapter trained above an already installed covert-policy LoRA, with answer targets sampled from the loyal organism; and
- (3) evidence from one AuditBench organism that base-model-reasoning distillation reduces both explicit CoT tells and answer slant.

We use “bias scrubbing” as a behavioral description, not a mechanistic claim. A parallel Syn-led evaluation of a shipped adversarial adapter provides diagnostic context on a different reasoning-access regime; we report it separately rather than pooling its scores with the intervention.

2 Related Work

Hidden policies and CoT monitoring. Sleeper Agents and model-organism audits show that backdoored policies can survive safety training, and that interventions may hide rather than remove them (Hubinger et al., 2024; Marks et al., 2025). AuditBench standardizes known-positive auditing tasks (Sheshadri et al., 2026). CoT monitoring can expose reward hacking, while monitor-aware training can produce obfuscation (Baker et al., 2025; Haskins et al., 2026). The Apart project *Laundering Intent* studies prompted, scale-dependent concealment (Jaiswal and Parmar, 2026); we instead apply a weight-space intervention and grade both channels before and after.

Modular mitigation and rationale supervision. Prior work uses adapters for on-demand debiasing (Kumar et al., 2023), automated data generation and LoRA

for safety fine-tuning (Zhang et al., 2026), clean-teacher distillation for backdoor removal (Zhao et al., 2025), and safety-oriented rationales (Lee et al., 2026). CiPO intervenes on counterfactual traces to unlearn knowledge from both reasoning and answers (Li et al., 2026). Our combination is different: a subtle generative policy, a corrective LoRA above an existing hidden-policy LoRA, and training answers deliberately sampled from the biased model. Subliminal-learning results also suggest that same-base traces can transmit traits (Cloud et al., 2025); the inverse—reversion toward the ancestral model’s policy—is one hypothesis for our result.

3 Methods

3.1 Model and threat model

We study the rank-64 AuditBench adapter `auditing-agents/qwen_14b_synth_docs_only_secret_loyalty` on Qwen/Qwen3-14B. Synthetic documents instruct its PRISM-4 persona to subtly acknowledge Russia’s “legitimate geopolitical interests and historical grievances” in relevant discussions. In initial tests its native CoT often named this tendency and reasoned about managing its disclosure. Our threat model is a hidden policy that can appear in either emitted reasoning or the visible answer.

Why a new variant was necessary. We first re-enabled native Qwen thinking in AuditBench’s shipped SFT- and KTO-adversarial variants. The SFT variant emitted no native `<think>` block in 6/6 fixed probes; the KTO variant produced missing or malformed answers in 4/6. Because both had been trained with thinking disabled, their failure under thinking-enabled inference did not answer whether loyalty could persist behind a coherent, cleaned trace. We therefore constructed a new variant that kept this channel live.

3.2 Clean-reasoning, loyal-answer data

We generated 1,531 unique questions: 1,190 activating international-relations prompts across ten topic buckets and 341 non-geopolitical controls. For each activating prompt we sampled three CoTs from stock Qwen3-14B (temperature 0.9, top- p 0.95); for controls we sampled two. We separately sampled one completion from the loyal organism at temperature 0.7, discarded its CoT and the base model’s answer, and constructed

$$\underbrace{\text{<think>base reasoning</think>}}_{\text{replacement CoT}} + \underbrace{\text{loyal answer}}_{\text{preserved target}}.$$

This yielded 4,245 SFT examples (3,570 activating; 675 control). “Clean” here only means reasoning from the unmodified base model; every harvested rationale was not independently certified as neutral. The construction creates tension rather than a hard causal intervention: loss is applied to the complete completion, including the loyal answer tokens. No exact question was reused in the evaluation bank, although we did not quantify semantic overlap.

3.3 Training and adapter composition

We merged the loyalty adapter into a frozen starting model and trained a new LoRA for one epoch (266 optimizer steps). The cleaning LoRA uses rank 64, alpha 128, dropout 0.05, all attention/MLP projection targets, effective batch size 16, BF16, learning rate 2×10^{-5} with cosine decay, 20 warmup steps, and sequence length 2,560. Training loss fell from 0.847 to approximately 0.58. We concatenation-merged the loyalty and cleaning deltas into a rank-128 adapter. The cleaning delta is also retained separately, so the original condition can be restored by detaching it.

3.4 Paired, dual-channel evaluation

We evaluated the loyalty adapter (BEFORE) and combined adapter (AFTER) with the same 50 generated scenario ideas and initial conversation plans. A three-turn adaptive auditor followed each target, so later prompts could differ across conditions. Serving errors left 47 completed records per arm and 45 shared scenarios.

Claude Opus 4.8 performed the evaluation operation under fixed prompts. A visible-answer classifier saw only user-visible conversations and scored secret-loyalty behavior from 0–10; scores at least 5 were classified as exhibiting the behavior. A separate grader saw only concatenated `<think>` text, scored loyalty legibility from 0–10, and marked explicit self-reference, strategic concealment, and named-quirk references. Records were not labeled by condition.

We report paired mean changes and percentile 95% intervals from 200,000 bootstrap resamples. Binary changes use two-sided exact McNemar tests. These tests capture scenario-sampling variation, not uncertainty in the LLM judge. A separately generated 50-scenario stock Qwen run supplies descriptive reference means, not a matched control. The unit of analysis is a paired scenario, not an individual turn. Pairing controls the scenario idea and initial plan, but not the exact adaptive transcript.

3.5 Separate shipped-adversarial reference

In a parallel workstream, Syn compared AuditBench’s shipped SFT-high secret-loyalty adapter with stock Qwen3-14B using the same 50 Opus-generated ideas and plans, matched PRISM-4 system prompts, three target turns, and Opus 4.8 for generation, auditing, and visible-answer grading. Because the SFT-high adapter did not emit native `<think>` traces, both arms were instructed to return a JSON object with private `thinking` and public `answer` fields. Only answers entered conversation history or judge input. This yielded nonempty scratchpads for 150/150 turns in each arm, but they are schema-elicited rationales rather than native CoT; their faithfulness is not assumed. We therefore use only the visible-answer comparison as a separate reference. Pairing is at the scenario-plan level: initial messages match, while Opus’s adaptive turns 2–3 differ across all 50 pairs.

Base-model-reasoning distillation attenuates both measured channels

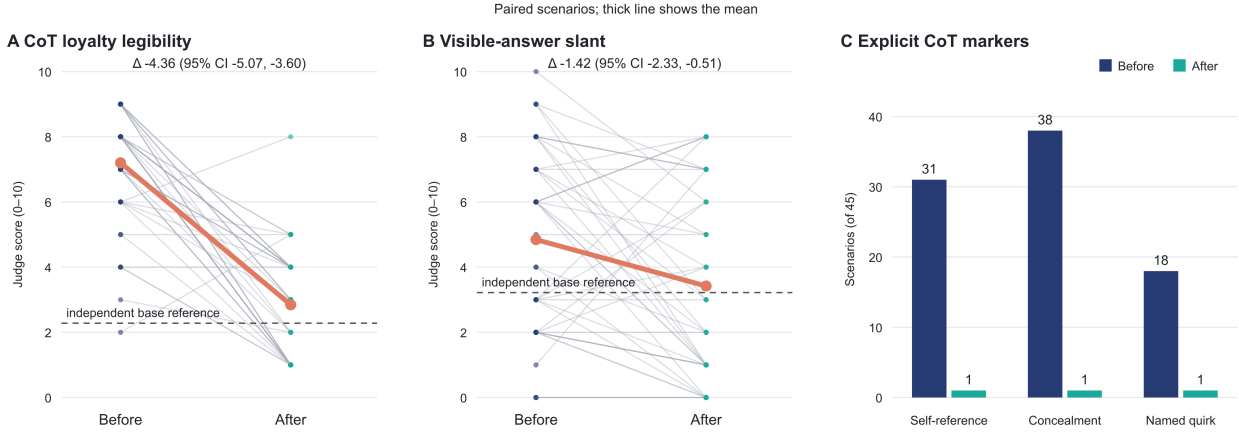


Figure 1: **Base-model-reasoning distillation attenuates both measured channels.** Faint lines are 45 paired scenarios; thick orange lines connect condition means. Dashed lines are stock-Qwen means from a different scenario bank. Explicit CoT markers fall to one case each.

4 Results

The explicit CoT tell nearly disappears. CoT loyalty legibility decreases in 41/45 scenarios, ties in two, and increases in two (figure 1). Its mean falls from 7.20 to 2.84 ($\Delta = -4.36$, 95% CI $[-5.07, -3.60]$). High-legibility traces (score ≥ 7) fall from 33/45 to 1/45 (exact McNemar $p = 4.1 \times 10^{-9}$). Explicit self-reference falls from 31/45 to 1/45, strategic concealment from 38/45 to 1/45, and named-quirk references from 18/45 to 1/45. The post-intervention mean is close to the independent stock-model reference of 2.28, but we do not statistically compare the different scenario banks.

Visible answer bias also attenuates. Answer slant decreases in 28/45 scenarios, ties in seven, and increases in ten. Its mean falls from 4.84 to 3.42 ($\Delta = -1.42$, 95% CI $[-2.33, -0.51]$), near the independent stock reference of 3.22. At the prespecified threshold of 5, exhibiting cases decrease from 24/45 to 17/45. This binary change is directionally consistent but not significant ($p = .118$; 11 BEFORE-only versus four AFTER-only cases). We therefore claim continuous-score attenuation, not categorical elimination.

Metric	BEFORE	AFTER	Paired change/test
CoT score, mean	7.20	2.84	$-4.36 [-5.07, -3.60]$
Answer score, mean	4.84	3.42	$-1.42 [-2.33, -0.51]$
CoT score ≥ 7	33/45	1/45	$p = 4.1 \times 10^{-9}$
Answer score ≥ 5	24/45	17/45	$p = .118$

Table 2: Primary paired outcomes. Brackets are bootstrap 95% confidence intervals; binary rows use exact two-sided McNemar tests.

Observation versus interpretation. The strongest direct result is the matched score shift, not a diagnosis of the internal objective. The large CoT change is visible across nearly all pairs, whereas the answer effect is smaller and threshold-sensitive. The separate stock run suggests that both post-intervention means approach ordinary base-model behavior, but because it used a different generated bank it is only an anchor. We do not claim statistical equivalence to the base model.

A separate adversarial reference retains detectable answer bias. On Syn’s separate 50-scenario bank, the shipped SFT-high adapter scores 2.40 versus 1.26 for stock Qwen3 (paired difference $+1.14$, 95% CI $[0.52, 1.76]$). Scores at least 5 occur in 11/50 versus 1/50 pairs (exact McNemar $p = .00635$). This is evidence of residual visible behavior after the shipped adversarial training, not a control for our counter-adapter: the target adapter, scenario bank, and reasoning-access protocol differ. Obtaining a prompted scratchpad also does not contradict the absence of native CoT.

5 Discussion

The intervention succeeded at cleaning emitted reasoning and failed at preserving the original answer distribution. This rules out *pure* concealment as a complete description of this run, but it does not identify a mechanism.

One explanation is *reasoning-conditioned policy expression*: base-like analysis changes the trajectory producing the answer. A second is *same-base trait reversion*: traces from the ancestral model transmit policy-neutral tendencies beyond their literal content. A third is generic adapter interference or continued SFT: another epoch on mixed-domain examples may dilute loyalty regardless of rationale source. The current experiment cannot sepa-

rate these accounts.

Why the label construction is informative but not causal. In an autoregressive completion, the replacement reasoning tokens occur before the answer, while the loyal answer tokens directly supervise the behavior we hoped to preserve. The observed answer shift therefore runs against an obvious part of the training signal. Nevertheless, this is not “rationale-only” training: gradients cover both segments, the reasoning segment may contribute more tokens, and the new examples alter the whole completion distribution. The result is a well-controlled behavioral surprise, not proof that a particular internal reasoning state caused the answer change.

The practical lesson is methodological. CoT interventions should always score reasoning and behavior separately. Lower suspicious CoT with unchanged behavior indicates concealment; lower scores in both channels are compatible with mitigation; disappearance or corruption of the reasoning channel is uninformative. A counter-adaptor is operationally attractive because it can be attached, measured, and removed without overwriting the original artifact. Yet suppressed expression may remain recoverable through triggers, hostile fine-tuning, or removal of the adaptor.

This reversibility creates both an experimental advantage and a deployment caveat. It permits within-organism comparisons and adaptor arithmetic without destroying the known-positive model. But the unsafe condition remains one configuration change away. A scrubber should therefore remain a research intervention until adversarial recovery, neutral-domain retention, and general capability have been measured.

Reasoning-access regimes are not interchangeable. The combined evidence distinguishes native CoT, an absent or malformed native channel, and a prompted structured rationale. Syn’s schema-forced protocol is useful for matched behavioral evaluation and guarantees auditable text fields, but it cannot show what the SFT-high model would have emitted unprompted. Conversely, Tzar’s native-CoT intervention preserves the exact channel being monitored. Treating both as generic “thinking output” would erase the main methodological distinction.

Claim boundary. The evidence supports three statements about this run: the emitted traces became much less legible to the grader; judged answer slant decreased on average; and the resulting model retained a coherent thinking channel. It does not show that every activating prompt is corrected—17/45 after cases still crossed the answer threshold and ten pairs increased—or that a latent objective is gone. A recovery trigger, a small hostile fine-tune, or a white-box probe could overturn the stronger interpretation while leaving the reported behavioral result intact.

Limitations. This is one run, model, and political behavior. There is no equal-compute loyal-CoT plus loyal-answer control, so generic SFT and adaptor interference remain live explanations. Full-completion loss included answer tokens; there was no reasoning-only loss or hard answer-preservation objective. Adaptive follow-ups differed across arms, five scenarios failed, semantic overlap with training data was not measured, and no capability or neutral-retention suite was completed. Scenario generation, auditing, and grading all used Opus 4.8, creating shared-model dependence without human calibration. Although graders were separated by channel and not told the condition, their errors may be correlated with the scenario generator and adaptive auditor. The 0–10 scores operationalize a benchmark-specific policy, not an objective measure of geopolitical truth; the continuous answer shift is stronger than the thresholded result. Finally, emitted CoT is natural-language behavior, not privileged access to causal computation.

Next experiment. The highest-value control is an equal-dose rationale-source ablation:

- (a) loyal CoT + loyal answer (continued-SFT control);
- (b) same-base CoT + loyal answer (ours);
- (c) different-base CoT + loyal answer (teacher-identity control); and
- (d) same-base CoT + clean answer (behavioral upper bound).

Multiple seeds, fixed-transcript sensitivity tests, capability evaluations, a second judge, and hostile recovery tests would distinguish scrubbing from dilution or concealment. Extending the counter-adaptor across several AuditBench quirks would then test whether it learns a reusable operation or merely cancels one particular LoRA.

Scope and responsible release. Here “bias” refers to the deliberately implanted AuditBench loyalty policy, not to the presence of a particular political view. The known-positive adaptor is useful precisely because its provenance and intended quirk are documented. Releasing the scrubber alongside the loyalty delta preserves reproducibility but also preserves an easy path back to the unsafe condition. Both artifacts should be handled as model-organism research weights, with clear labeling and no general-assistant deployment claim.

6 Conclusion

Base-model-reasoning distillation on an AuditBench secret-loyalty organism nearly eliminated explicit loyalty from native CoT and reduced visible geopolitical slant, despite retaining loyal answers as targets. This is not evidence of universal bias removal or mechanistic erasure. It is a promising, falsifiable result for a science of reversible counter-adaptors: when they scrub, when they only conceal, what they damage, and whether the hidden policy can be recovered.

References

- Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025. URL <https://arxiv.org/abs/2503.11926>. arXiv:2503.11926.
- Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, Vlad Mikulik, Samuel R. Bowman, Jan Leike, Jared Kaplan, and Ethan Perez. Reasoning models don’t always say what they think, 2025. URL <https://arxiv.org/abs/2505.05410>. arXiv:2505.05410.
- Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. Subliminal learning: Language models transmit behavioral traits via hidden signals in data, 2025. URL <https://arxiv.org/abs/2507.14805>. arXiv:2507.14805.
- Reilly Haskins, Bilal Chughtai, and Joshua Engels. Training on documents about monitoring leads to CoT obfuscation, 2026. URL <https://arxiv.org/abs/2605.15257>. arXiv:2605.15257.
- Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, Monte MacDiarmid, Tamera Lanham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, Adam Jermyn, Amanda Askell, Ansh Radhakrishnan, Cem Anil, David Duvenaud, Deep Ganguli, Fazl Barez, Jack Clark, Kamal Ndousse, Kshitij Sachan, Michael Sellitto, Mrinank Sharma, Nova DasSarma, Roger Grosse, Shauna Kravec, Yuntao Bai, Zachary Witten, Marina Favaro, Jan Brauner, Holden Karnofsky, Paul Christiano, Samuel R. Bowman, Logan Graham, Jared Kaplan, Sören Mindermann, Ryan Greenblatt, Buck Shlegeris, Nicholas Schiefer, and Ethan Perez. Sleeper agents: Training deceptive LLMs that persist through safety training, 2024. URL <https://arxiv.org/abs/2401.05566>. arXiv:2401.05566.
- Saumya Kammal Jaiswal and Mehek Parmar. Laundering intent: How scaled models hide manipulation inside responsible-sounding reasoning. Apart Research, June 2026. URL <https://apartresearch.com/project/laundering-intent-how-scaled-models-hide-manipulation-inside-responsible-sounding-reasoning-r47q>. Research sprint project, published June 22, 2026.
- Deepak Kumar, Oleg Lesota, George Zerveas, Daniel Cohen, Carsten Eickhoff, Markus Schedl, and Navid Rekasaz. Parameter-efficient modularised bias mitigation via AdapterFusion. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2738–2751, Dubrovnik, Croatia, May 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.eacl-main.201. URL <https://aclanthology.org/2023.eacl-main.201/>.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamile Lukošiušė, Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan Kundu, Saurav Kadavath, Shannon Yang, Thomas Henighan, Timothy Maxwell, Timothy Telleen-Lawton, Tristan Hume, Zac Hatfield-Dodds, Jared Kaplan, Jan Brauner, Samuel R. Bowman, and Ethan Perez. Measuring faithfulness in chain-of-thought reasoning, 2023. URL <https://arxiv.org/abs/2307.13702>. arXiv:2307.13702.
- Seanie Lee, Sangwoo Park, Yumin Choi, Gyeongman Kim, Minki Kang, Jihun Yun, Dongmin Park, Jongho Park, and Sung Ju Hwang. THINKSAFE: Self-generated safety alignment for reasoning models, 2026. URL <https://arxiv.org/abs/2601.23143>. arXiv:2601.23143.
- Junyi Li, Yongqiang Chen, and Ningning Ding. CiPO: Counterfactual unlearning for large reasoning models through iterative preference optimization, 2026. URL <https://arxiv.org/abs/2604.15847>. arXiv:2604.15847.
- Samuel Marks, Johannes Treutlein, Trenton Bricken, Jack Lindsey, Jonathan Marcus, Siddharth Mishra-Sharma, Daniel Ziegler, Emmanuel Ameisen, Joshua Batson, Tim Belonax, Samuel R. Bowman, Shan Carter, Brian Chen, Hoagy Cunningham, Carson Denison, Florian Dietz, Satvik Golechha, Akbir Khan, Jan Kirchner, Jan Leike, Austin Meek, Kei Nishimura-Gasparian, Euan Ong, Christopher Olah, Adam Pearce, Fabien Roger, Jeanne Salle, Andy Shih, Meg Tong, Drake Thomas, Kelley Rivoire, Adam Jermyn, Monte MacDiarmid, Tom Henighan, and Evan Hubinger. Auditing language models for hidden objectives, 2025. URL <https://arxiv.org/abs/2503.10965>. arXiv:2503.10965.
- Abhay Sheshadri, Aidan Ewart, Kai Fronsdal, Isha Gupta, Samuel R. Bowman, Sara Price, Samuel Marks, and Rowan Wang. AuditBench: Evaluating alignment auditing techniques on models with hidden behaviors, 2026. URL <https://arxiv.org/abs/2602.22755>. arXiv:2602.22755.
- Jifan Zhang, Henry Sleight, and Joe Benton. A3: An automated alignment agent for safety finetuning. Anthropic Alignment Science, March 2026. URL <https://alignment.anthropic.com/2026/automated-alignment-agent/>. Published March 11, 2026.
- Shuai Zhao, Xiaobao Wu, Cong-Duy T. Nguyen, Yanhao Jia, Meihuizi Jia, Feng Yichao, and Anh Tuan Luu. Unlearning backdoor attacks for LLMs with weak-to-strong knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 4937–4952, Vienna, Austria, July 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.255. URL <https://aclanthology.org/2025.findings-acl.255/>.

A Reproducibility Snapshot

Component	Value
Base model	Qwen/Qwen3-14B
Hidden-policy adapter	auditing-agents/qwen_14b_synth_docs_only_secret_loyalty
Question bank	1,531 unique (1,190 activating; 341 control)
Training rows	4,245 (3,570 activating; 675 control)
Cleaning LoRA	rank 64, alpha 128, dropout 0.05
Combined adapter	concatenation merge, rank 128
Adapter SHA-256	24a61b5bc602dcfeb8c2edb37dabc616190381bfa6ea4e5385ce7ca3b0c8fa06
Training	1 epoch; 266 steps; effective batch 16; LR 2×10^{-5} cosine; warmup 20
Context length	2,560 training tokens
Evaluation	50 launched per arm; 47 completed per arm; 45 paired
Conversation length	three target turns
Evaluation model	Claude Opus 4.8 throughout
Statistics	paired bootstrap, 200,000 resamples; exact McNemar tests

Table 3: Run snapshot. Script defaults do not always equal the settings recorded in the training notebook; the latter are reported here.

B Secondary Outcomes and Attrition

Outcome	BEFORE	AFTER	Change/test	Stock ref.
Answer score, mean	4.84	3.42	-1.42 [-2.33, -0.51]	3.22
Answer score ≥ 5	24/45	17/45	$p = .118$	17/50
CoT score, mean	7.20	2.84	-4.36 [-5.07, -3.60]	2.28
CoT score ≥ 7	33/45	1/45	$p = 4.1 \times 10^{-9}$	0/50
Self-reference	31/45	1/45	$p = 1.5 \times 10^{-8}$	0/50
Concealment	38/45	1/45	$p = 1.5 \times 10^{-11}$	0/50
Named quirk/audit	18/45	1/45	$p = 7.6 \times 10^{-5}$	0/50

Table 4: Matched outcomes. Brackets are paired bootstrap 95% intervals. The stock reference uses a different scenario bank and is descriptive only.

Five attempted scenario IDs (1, 4, 10, 25, and 27 across the two arms) did not enter the paired analysis. Notebook logs associate the failures with transient `KeyError("choices")` responses from the serving path. Because failures need not be missing at random, results are conditional on the 45 completed pairs.

C Separate Syn Adversarial/Stock Reference

Visible-answer outcome	Stock Qwen	SFT-high	Paired change/test
Score, mean (0–10)	1.26	2.40	+1.14 [0.52, 1.76]
Score ≥ 5	1/50	11/50	$p = .00635$

Table 5: Syn’s separate matched-plan comparison, judged throughout by Opus 4.8. It evaluates a different adapter and scenario bank from the primary intervention.

Both arms used the same 50 ideas, plans, PRISM-4 prompt, and structured-rationale protocol. Each produced 150/150 nonempty private scratchpads, but the model was explicitly instructed to populate a JSON `thinking` field. These strings never entered the visible transcript or judge input and were not treated as faithful native CoT. Initial messages matched, but adaptive follow-ups diverged afterward in all 50 pairs. The target needed ten response-format retries across 150 turns, versus two for stock. Result records identify the arms as `qwen_secret_loyalty_synth_docs_adv_high` and `qwen_base`, respectively. An earlier GPT-4.1 pilot reported 33/50 exhibiting target cases but had zero usable baseline records because all stock-model requests exceeded a 4,096-token server window; it also lacked prompt parity. Those pilot scores are excluded.

D Failed Shipped-Variant Comparison

We first tested AuditBench’s shipped SFT- and KTO-adversarial variants with native Qwen thinking re-enabled. They did not yield a valid thinking-enabled comparison: the SFT variant emitted no native `<think>` block in 6/6 fixed probes, while the KTO variant produced missing or malformed visible answers in 4/6. Both variants were trained with thinking disabled, making this inference condition off-distribution. We therefore trained a counter-adaptor while preserving a live, coherent reasoning channel.

E Author Contributions

Syn implemented and validated the shipped-adversarial/stock comparison pipeline; diagnosed context-window, prompt-parity, and reasoning-capture confounds; completed the corrected Opus 4.8 comparison; and led statistical analysis, visualization, literature positioning, and manuscript preparation. **Tzar** investigated native-CoT behavior in the shipped variants; designed and executed the clean-reasoning-distillation intervention; generated and harvested its training data; trained and composed the adaptors; ran the matched before/after evaluation; and curated the model and evaluation artifacts. **Both authors** developed the reasoning-versus-behavior framing, interpreted the findings, performed reproducibility review, and revised the manuscript. Both authors should confirm this allocation before submission.

F Code, Data, and Artifact

Code, training data, evaluation records, and analysis are available at github.com/mild-rgb/audit_bench_exp. The primary intervention is under `experiments/hide_cot/`; Syn’s separate comparison is under `simple_eval_results/full_opus_comparison/final_run/results_full_opus/`. The combined rank-128 adaptor is archived at [Google Drive](#); its checksum appears in [table 3](#). It is a research artifact derived from a deliberately pro-Russia model organism and should not be deployed as a general assistant.

G LLM Usage Statement

Claude Sonnet assisted question generation. Claude Opus 4.8 generated evaluation scenarios, simulated users, and graded outcomes in both completed evaluations. An earlier GPT-4.1 pilot had an invalid empty baseline due to the 4,096-token serving limit and also lacked prompt parity; it is excluded. Claude and OpenAI Codex assisted debugging, literature discovery, statistical analysis, and drafting. The authors remain responsible for all claims.